

Billie Gene

Billie Gene is an AI-assisted vaccine design workflow that helps researchers move from a pathogen sequence to a ranked vaccine candidate dossier.

The app takes a viral/pathogen sequence as input, identifies likely surface proteins, annotates their function and structural properties, predicts candidate binding epitopes, designs a conceptual mRNA construct encoding selected epitopes, and screens for pharmacogenomic risk signals that may affect population-level safety.

Billie Gene is built as a guided stepper interface: each stage of the pipeline lives in its own tab so users always know where they are in the vaccine design workflow.

Hackathon / research-use disclaimer: Billie Gene is a non-clinical prototype for computational exploration and educational demonstration. It does not generate clinically validated vaccine products, therapeutic recommendations, or wet-lab-ready protocols.

Problem

Modern vaccine design requires researchers to connect many disconnected tasks:

1. Parse a pathogen genome or protein sequence
2. Identify surface-exposed proteins likely to trigger immune response
3. Annotate protein function, structure, stability, and developability
4. Predict high-quality epitopes
5. Design an mRNA sequence encoding selected antigenic regions
6. Estimate population-level pharmacogenomic risk
7. Produce a clear report that explains the rationale

Billie Gene turns this into a single interactive workflow.

Core Pipeline

Pathogen sequence

↓

Identify surface protein

↓

Annotate function + developability
↓
Predict best binding epitopes
↓
Design conceptual mRNA construct
↓
Pharmacogenomic risk screen
↓
Generate vaccine candidate report

App Flow: Stepper Tabs

Step 01 — Sequence Input

Goal: Let the user upload or paste a pathogen sequence.

Supported demo inputs:

- FASTA sequence
- Raw nucleotide sequence
- Protein accession ID
- Example viral variant sequence
- Reference + mutant sequence pair

UI elements:

- Upload card
- Paste FASTA input
- Pathogen type dropdown
- Reference strain selector
- “Run analysis” button
- Sequence validation panel

Outputs:

- Sequence length
 - GC content
 - ORF count
 - Predicted coding regions
 - Input confidence score
-

Step 02 — Surface Protein Discovery

Goal: Identify which encoded proteins are likely surface-exposed and relevant for vaccine targeting.

This is the screen that needs the strongest visual.

Instead of showing only a list of proteins, Billie Gene should show a **protein segmentation map** inspired by genome browser tracks and splicing diagrams.

Recommended visual: “Surface Protein Track Viewer”

Show the protein as a long horizontal ribbon divided into colored regions:

N-term Signal Peptide Ectodomain Transmembrane Cytoplasmic Tail
|-----|-----|-----|-----|-----|

Then overlay:

- Predicted domains
- Surface-exposed regions
- Transmembrane regions
- Glycosylation sites
- Mutation hotspots
- Candidate epitope windows
- Confidence bars
- Variant differences

Why this works: It gives the “AlphaGenome-like” feeling of tracks, but applies it to viral protein biology instead of human regulatory genomics. AlphaGenome-style visualizations are useful inspiration because the model compares functional effects across genomic regions, but Billie Gene’s surface-protein screen should focus on **protein domains, structure, and epitope accessibility**.

UI concept:

Surface Protein Candidate: Spike-like Glycoprotein

[Signal peptide][Surface domain A][Receptor binding region][Fusion peptide][TM][Tail]
 ↑ ↑ ↑ ↑
cleavage site epitope window mutation cluster membrane anchor

Surface Exposure Score: 0.91

Vaccine Targetability: High

Outputs:

- Top surface protein candidates

- Surface exposure score
- Signal peptide prediction
- Transmembrane domain prediction
- Extracellular domain boundaries
- Structural model availability
- Vaccine targetability score

Step 03 — Function Annotation

Goal: Explain what the protein does and whether it is a good vaccine target.

This tab should combine biological annotation with practical developability metrics.

Suggested metrics:

Metric	What it means	Why it matters
Surface exposure	How accessible the protein is to antibodies	Higher exposure usually means better immune visibility
Functional importance	Whether the protein is involved in entry, attachment, fusion, or replication	Essential proteins are stronger vaccine targets
Conservation score	How stable the region is across variants	Conserved regions reduce immune escape risk
Mutation density	How often the region changes across variants	High mutation density may reduce durability
Predicted solubility	Whether the antigenic region is likely to express well	Useful for manufacturability
Structural stability	Whether the region is likely to fold reliably	Helps avoid unstable constructs
Binding affinity estimate	Predicted immune-binding strength	Helps rank epitopes
Glycosylation shielding	Whether sugars may hide the epitope	Shielded regions may be less effective
Toxicity / allergenicity flag	Whether the region resembles risky motifs	Helps filter candidates

Population coverage	How broadly HLA-binding predictions cover populations	Helps assess vaccine breadth
---------------------	---	------------------------------

Composite score:

Billie Gene Target Score =
25% surface exposure
+ 20% conservation
+ 20% predicted binding strength
+ 15% structural stability
+ 10% manufacturability
+ 10% population coverage
- risk penalties

Example output:

Function Annotation Summary

Protein: Surface glycoprotein candidate
Likely role: Host-cell attachment / entry
Surface exposure: High
Conservation: Medium-high
Predicted stability: Medium
Glycosylation shielding: Moderate
Overall Billie Gene Target Score: 82 / 100
Recommendation: Advance to epitope prediction

Step 04 — Epitope Prediction

Goal: Identify and rank the best candidate epitopes.

UI elements:

- Epitope table
- Protein track overlay
- Binding affinity heatmap
- Population coverage chart
- Risk filter toggles

Epitope ranking metrics:

Metric	Description
--------	-------------

MHC-I binding score	Potential CD8 T-cell presentation
MHC-II binding score	Potential CD4 T-cell presentation
B-cell epitope likelihood	Antibody recognition potential
Surface accessibility	Whether the epitope is exposed
Conservation	Whether it remains stable across variants
Variant escape risk	Whether mutations frequently occur there
Manufacturability	Whether sequence is easy to encode/express
Safety filters	Similarity to human proteins, allergenicity, toxicity flags

Recommended visual:

A protein ribbon with epitope windows:

Protein length: 1 ————— 1273

Epitope A: 
 Epitope B: 
 Epitope C: 
 Mutation hotspot: 
 Glycosylation sites: 

Each epitope card should show:

Epitope Candidate BG-EPI-003

Region: 412–427

Binding score: 94 / 100

Surface exposure: High

Conservation: 88%

Population coverage: 74%

Escape risk: Low

Decision: Include in mRNA construct

Step 05 — mRNA Construct Design

Goal: Generate a **conceptual** mRNA construct layout encoding the selected epitopes.

For the hackathon demo, avoid presenting this as a wet-lab-ready sequence generator. Instead, frame it as a **construct planner**.

Construct visual:

5' Cap | 5' UTR | Signal peptide | Epitope 1 | Linker | Epitope 2 | Linker | Epitope 3 | 3' UTR | Poly(A)

UI elements:

- Construct diagram
- Selected epitopes
- Linker strategy
- Codon optimization status
- GC balance
- Repeats / motif warnings
- Translation preview
- Export disabled or marked “simulation only” for hackathon safety

Metrics:

Metric	Purpose
GC content	Avoid sequences that are too GC-rich or too AT-rich
Codon adaptation score	Estimate host expression compatibility
Repeat risk	Avoid unstable repeated regions
Secondary structure risk	Identify problematic RNA folding
Linker compatibility	Ensure epitopes remain separated
Construct length	Keep construct within practical bounds
Translation integrity	Ensure no premature stop codons in demo translation
Manufacturability score	Composite score for construct quality

Example output:

Construct Candidate: BG-mRNA-001

Selected epitopes: BG-EPI-001, BG-EPI-003, BG-EPI-006

Construct length: 1,248 nt

GC content: 53%
Translation integrity: Pass
Repeat risk: Low
Secondary structure risk: Medium
Manufacturability score: 78 / 100

Step 06 — Pharmacogenomic Risk Screen

Goal: Estimate whether certain populations or genetic backgrounds may have higher risk signals.

This should be framed carefully as **risk exploration**, not diagnosis.

Inputs:

- Candidate epitopes
- Adjuvant/construct assumptions
- HLA-binding profile
- Population allele frequencies
- Known immune-risk markers, when available

Outputs:

- Population coverage
- HLA representation
- Potential hyper-response flags
- Potential low-response groups
- Similarity-to-human-protein warning
- Risk confidence level

Visuals:

- Population coverage map
- HLA allele coverage table
- Risk matrix
- Confidence meter

Example UI card:

Pharmacogenomic Screen

Predicted population coverage: 81%
Low-response risk: Moderate
Autoimmunity similarity flag: None detected in demo database
Confidence: Medium

Recommendation:

Proceed as a computational candidate, but require experimental validation.

Step 07 — Testing Hypotheses / Assay Planning

Goal: Let the user ask targeted research questions and produce a safe computational testing plan.

Since your examples are about missense/splicing variants, adapt the concept to viral vaccine design:

Example vaccine-focused prompts:

Compare the reference and variant surface proteins. Which mutations are most likely to reduce antibody recognition?

Generate competing hypotheses for why this viral variant may escape immune binding.

Visualize the surface protein structure and color candidate escape mutations in red.

Compare epitope conservation between the reference strain and the submitted variant.

Create a testing plan to validate whether Epitope BG-EPI-003 remains immunogenic across variants.

Safe demo command example:

```
billie-gene analyze-variant \  
--reference examples/reference_surface_protein.fasta \  
--variant examples/variant_surface_protein.fasta \  
--mode epitope-escape \  
--visualize structure \  
--output reports/variant_escape_report.json
```

Example output:

Competing hypotheses

1. Mutation cluster near Epitope BG-EPI-003 may reduce antibody recognition.
2. Glycosylation-adjacent changes may shield the receptor-binding region.
3. The epitope remains structurally intact, but HLA presentation may be reduced.

Recommended validation path

- Compare reference vs variant epitope conservation
- Visualize structural position of mutations
- Re-score epitope binding predictions

- Flag candidates that lose surface accessibility or binding strength

For structural visuals, PyMOL is a reasonable way to render protein structures, and AlphaFold DB can provide predicted protein structures when available. AlphaFold DB is a database of AI-predicted protein structures, while PyMOL is commonly used for molecular visualization.

Example PyMOL-style visualization command for the UI, not necessarily executed in-browser:

```
load surface_protein_model.pdb
select variant_sites, resi 417+484+501
color red, variant_sites
show spheres, variant_sites
zoom variant_sites
```

Suggested UI Layout

Left Sidebar

Billie Gene

- 01 Sequence Input
- 02 Surface Protein
- 03 Function Annotation
- 04 Epitope Prediction
- 05 mRNA Construct
- 06 PGx Risk Screen
- 07 Final Report

Main Stepper

[1 Input] → [2 Surface Protein] → [3 Annotation] → [4 Epitopes] → [5 mRNA] → [6 PGx] → [7 Report]

Each tab should have:

- Step title
 - What this step does
 - Input panel
 - Processing state
 - Visualization panel
 - Metrics panel
 - “Continue” button
 - “Explain this result” assistant card
-

Key Visuals to Build

1. Surface Protein Track Viewer

This is the most important screen.

Show a protein split into domains:

Signal peptide | NTD | RBD | Fusion peptide | HR1 | HR2 | TM | Tail

Overlay:

- Surface exposure
- Mutation sites
- Epitope windows
- Glycosylation sites
- Confidence track

This gives the user the “splicing/track” feeling you wanted without incorrectly relying on a human-genomics model for viral antigen discovery.

2. Structure Viewer

Use a 3D protein viewer or static rendered mock.

Features:

- Protein backbone
 - Candidate epitopes in blue
 - Mutations in red
 - Glycosylation sites in purple
 - High-confidence target regions highlighted
-

3. Epitope Heatmap

Rows = candidate epitopes

Columns = scoring dimensions

	Binding	Surface	Conserved	Stable	Safe	Coverage
BG-EPI-001	92	88	90	76	84	79
BG-EPI-002	81	94	72	80	88	85

4. mRNA Construct Diagram

5' Cap — 5' UTR — Signal — EPI-001 — Linker — EPI-003 — Linker — EPI-006 — 3' UTR — Poly(A)

Each segment should be clickable.

5. Final Dossier

A report-style screen with:

- Candidate summary
 - Protein target rationale
 - Top epitopes
 - mRNA construct layout
 - PGx risk summary
 - Confidence score
 - Limitations
 - Suggested next computational validations
-

Example Scoring System

Billie Gene Candidate Score: 84 / 100

Surface Target Score: 91
Function Relevance Score: 86
Epitope Quality Score: 88
Conservation Score: 82
Manufacturability Score: 78
PGx Risk Score: 74
Overall Confidence: Medium-high

Interpretation:

This candidate targets a highly exposed surface protein with strong predicted immune-binding regions and moderate manufacturability. Some regions show variant sensitivity, so Billie Gene recommends including conserved epitopes and excluding mutation-dense windows.

Example README Installation Section

Getting Started

Prerequisites

- Node.js 18+
- Python 3.10+
- npm or pnpm

Install

```
```bash
git clone https://github.com/your-org/billie-gene.git
cd billie-gene
pnpm install
```

## Run the app

```
pnpm dev
```

## Run demo pipeline

```
python scripts/run_demo_pipeline.py \
--input examples/demo_pathogen.fasta \
--output reports/demo_report.json
```

---

## ## Example README Tech Stack

```
```md
```

Tech Stack

- Frontend: Next.js / React
- Styling: Tailwind CSS
- Visualization: D3.js, Mol* or NGL Viewer for protein structure
- Backend: Python / FastAPI
- Bioinformatics utilities: Biopython-style parsing, protein-domain annotation, epitope scoring adapters
- Reports: JSON + downloadable PDF dossier

For browser-based protein visualization, I'd use **Mol*** or **NGL Viewer** in the actual app, and use PyMOL-style output as an export or demo command.

Example Final Report

Vaccine Candidate Report

Candidate ID

BG-CANDIDATE-001

Target Protein

Predicted surface glycoprotein

Rationale

The selected protein is predicted to be surface-exposed, functionally relevant to host-cell entry, and contains multiple conserved epitope windows with strong binding predictions.

Selected Epitopes

ID	Region	Binding Score	Conservation	Surface Exposure	Decision
BG-EPI-001	112–128	91	88%	High	Include
BG-EPI-003	411–427	95	84%	High	Include
BG-EPI-006	701–718	87	91%	Medium	Include

Construct Summary

The conceptual mRNA construct includes three selected epitopes separated by linker regions and wrapped in standard mRNA construct components.

Risk Summary

Population coverage is predicted to be broad, but several HLA groups show lower predicted presentation. No high-confidence safety flags were detected in the demo screen.

Recommendation

Advance this candidate for deeper computational validation and experimental review.

Suggested One-Liner

Billie Gene is an AI vaccine design copilot that turns pathogen sequences into explainable, ranked vaccine candidates.

Suggested Tagline

From sequence to vaccine strategy — one guided step at a time.